

**SCHOOL OF
COMPUTING
CHENNAI**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSE304 Embedded Systems Laboratory

Activity Sheet #5

Activity Name: DAC Module Interfacing

Name of Student:	C S DEERAJ
Registration No.:	CH.SC.U4CSE24106
Activity Date:	18/08/2026
Submission Date:	18/08/2026

Assessment Rubrics:

Component	Maximum Marks	Marks Allotted
Program	3	
Output	4	
Viva	2	
Timely Submission	1	
Total	10	

Faculty In-charge

Write an embedded C program for ARM architecture to drive DAC module. Debug and verify the program using IDE on a development computer system.

Embedded C code:

```
#include <LPC21xx.H>

#define DACR (*((volatile unsigned long *)0xE006C000))

// DAC register address

void delay(unsigned int t) {
//CH.SC.U4CSE24106
    unsigned int i;
    for(i=0; i<t; i++);
}

int main(void) {
    unsigned int value;
    PINSEL1 |= 0x00080000;
    while(1)
    {
        for(value=0; value<=1023; value++)
        {
            DACR = (value << 6);
            delay(50);
        }
        for(value=1023; value>0; value--)
        {
            DACR = (value << 6);
            delay(50);
        }
    }
}
```

Evaluation Mode Registers

C:\Users\CH.SC.U4CSE24106\Desktop\CH.SC.U4CSE24106 Embedded Systems\Lab Exercise 5 - DAC module\DAC module.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers Disassembly

Register	Value
Current	
R0	0x40000060
R1	0x40000060
R2	0x40000060
R3	0x40000060
R4	0x00000000
R5	0x40000000
R6	0x00000000
R7	0x00000000
R8	0x00000000
R9	0x00000000
R10	0x0000023C
R11	0x00000000
R12	0x000001B4
R13 (SP)	0x40000460
R14 (LR)	0x000000C0
R15 (PC)	0x000001B4
CPSR	0x600000D3
SPSR	0x00000000
User/System	
Fast Interrupt	
Interrupt	
Supervisor	
Abort	
Undefined	
Internal	
PC \$	0x000001B4
Mode	Supervisor
States	866
Sec	0.00007217

```
0x000001B0 E12FFF1E BX R14
12:
13:      while(1)
->0x000001B4 E59F0064 LDR    R0,[PC,#0x0064]
0x000001B8 E5900004 LDR    R0,[R0,#0x0004]
0x000001BC E3800702 ORR    R0,R0,#0x00080000
.....
```

DAC module code.c

```
1  #include <LPC21xx.H>
2  #define DACR (*(volatile unsigned long *)0xE006C000) // DAC register address
3  void delay(unsigned int t)
4  {
5      unsigned int i;
6      for(i=0; i<t; i++);
7  }
8  int main(void)
9  {
10     unsigned int value;
11     PINSEL1 |= 0x00080000;
12
13     while(1)
14     {
15         for(value=0; value<=1023; value++)
16         {
17             DACR = (value << 6);
18             delay(50);
19         }
20         for(value=1023; value>0; value--)
21         {
22             DACR = (value << 6);
23             delay(50);
24         }
25     }
26 }
```

Command

Running with Code Size Limit: 32K
Load "C:\\Users\\CH.SC.U4CSE24106\\Desktop\\CH.SC.U4CSE24106 Embedded Systems\\Lab Exercise 5 - DAC module\\Objects\\D
*** Restricted Version with 32768 Byte Code Size Limit
*** Currently used: 572 Bytes (1%)

Call Stack - Locals

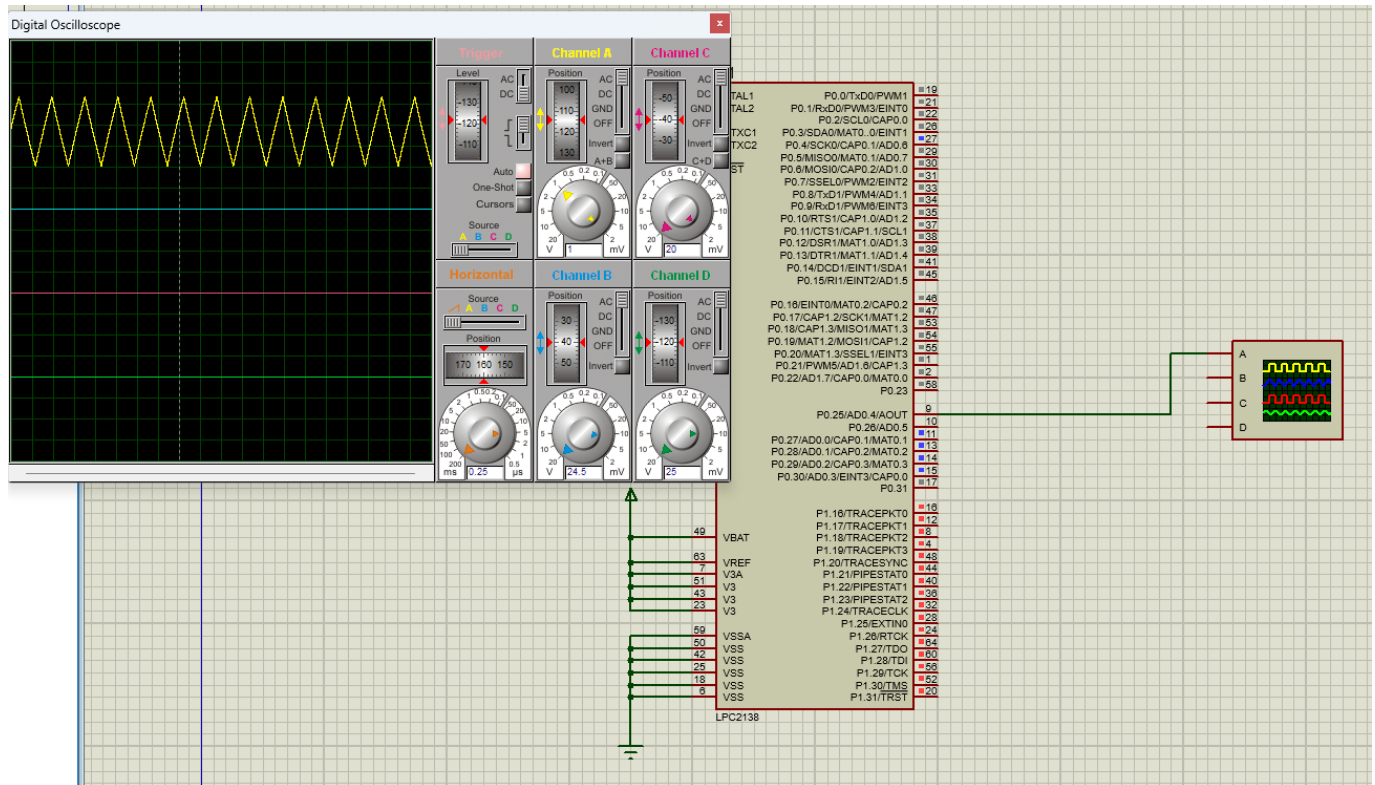
Name	Location/V...	Type
main	0x000001B4	int f0
v...	0x40000060	auto - unsig...

ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE DEFINE DIR Display Enter EVALUATE

Call Stack + Locals Memory 1

Real-Time Agent: Target Reset Simulation

Output:



Note from C S Deeraj to Embedded Systems Faculty:

So what i have interpreted and inferred from this experiment is that

The oscilloscope waveform shows a smooth triangular wave, confirming that the LPC2138's built-in DAC is working correctly. As the program increases the digital value from 0 to 1023, the analog output voltage rises steadily toward 3.3V; as the value decreases back to 0, the voltage falls back down. This linear rise and fall shows that the DAC is accurately converting digital values into a proportional analog voltage.

This confirms that the DAC was correctly configured (PINSEL1 for AOUT function, and the digital value properly aligned using value $\ll 6$), and that the digital-to-analog conversion was successfully verified using Keil μ Vision and Proteus.